S M MAHAMUDUL HASAN

AI / ML ENGINEER | LLM APPLICATIONS, GENAI & MACHINE LEARNING | PYTHON, PYTORCH, FASTAPI

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PROFESSIONAL SUMMARY

AI/ML engineer with 3+ years of experience building LLM applications, machine learning models and the backend services that run them, an MSc in AI and Robotics (2026) and three peer-reviewed IEEE and Springer papers. Ships GenAI end to end, including a bilingual LLM chat service with 9 FastAPI endpoints, streaming, session memory, Docker Compose and pytest suites. Takes classical and deep learning models from prototype to production, from Bangla BERT at 0.96 F1 to leakage-aware clinical ML research.

TECHNICAL SKILLS

LLMs and GenAI: LLM apps, chatbots, LLM APIs (GPT, Llama 3.2/Ollama), prompt and context engineering, RAG
Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, BERT fine-tuning, LSTM, CNN, feature engineering
NLP and Evaluation: Text classification, sentiment analysis, information extraction, Word2Vec, leakage-aware evaluation
AI Backend: Python, FastAPI, Flask, REST APIs, validated schemas, SQL, Pandas, NumPy, TypeScript
Deployment and Tools: Docker Compose, pytest, health checks, GitHub Actions, GitLab CI/CD, Azure DevOps, Git, Claude Code

AI AND ML ENGINEERING EXPERIENCE

Applied AI Engineer | BritAI Labs | Jan 2026 - Present

London, UK | Fixed-term contract

- Delivered a private bilingual LLM chat service with 9 REST endpoints and real-time NDJSON token streaming by building FastAPI routes on Llama 3.2 via Ollama with validated request/response schemas and Docker Compose.
- Kept multi-turn chats coherent with a 20-message context window per session by building session memory, Bangla/English system prompts and JSON/Markdown export.
- Made every release testable and monitorable without a running model by writing pytest unit and API tests against a fake LLM client and adding liveness and model-readiness health checks.

Application Engineer | All Generation Tech | Jun 2024 - Sep 2024

Remote, United States | Contract

- Improved system efficiency by 20% by integrating APIs and JSON data flows between client systems.
- Cut deployment time by 30% by working with the wider tech team to turn complex user stories into Unqork back ends.

Pricing Software Engineer | Ark Solutions | Jun 2022 - Sep 2023

Remote, Australia | Full-time

- Reduced manual effort by 95% by automating SAP data migration and multi-source data integrations.
- Cut pricing-processing time by 25% and operational errors by 15% by optimising Groovy rules and leading unit testing.

Machine Learning and NLP Research Apprentice | HAB Lab | Feb 2020 - May 2022

Sylhet, Bangladesh | On-site | HCI, AI and Blockchain Laboratory

- Reached 0.96 F1 on Bangla fake-news detection by labelling 11,434 news articles, benchmarking five models up to fine-tuned Bangla BERT, and serving predictions through a Flask API to an app adopted by a media agency.
- Built and evaluated ML and NLP models for classification, sentiment and information extraction with LSTM and BERT.
- Led a five-member research team to three published studies by designing benchmarks against strong baselines.

SELECTED PROJECTS

Local LLM Chat API | Private Bangla/English chat on Ollama: FastAPI, NDJSON streaming, session memory, Docker Compose, pytest
Bangla Fake News Detection | 11,434 labelled articles; five models benchmarked up to fine-tuned Bangla BERT at 0.96 F1
Phishing Website Detection | Compared classical ML classifiers on URL and page features; published in Springer Nature LNEE

SELECTED PUBLICATIONS

Is Near-Perfect Dengue Prediction Real? A Leakage-Aware Evaluation of Machine Learning on a Public Clinical Dataset from Bangladesh
Under review | IEEE conference

A Benchmark of Machine Learning and Deep Learning Algorithms for Detecting Fake News in Bangla Language
Second author | IEEE STI 2022 | Code

Determining the Most Effective Machine Learning Techniques for Detecting Phishing Websites
First author | Springer Nature, LNEE vol. 925, 2022 | Code

Can Machine Learning Techniques Predict Prostate Cancer Accurately? The Fact and Remedy
First author | IEEE, 2021 | Code

EDUCATION

MSc in Artificial Intelligence and Robotics | University of Hertfordshire | Hatfield, UK | Sep 2024 - Sep 2026
BSc in Computer Science and Engineering | Shahjalal University of Science and Technology | Sylhet, Bangladesh | Jan 2017 - Dec 2022

AWARDS

Top 5 of 400+ teams, National Hackathon on Frontier Technology, Government of Bangladesh (2021) | Top 10, SigmaHacks (2022)